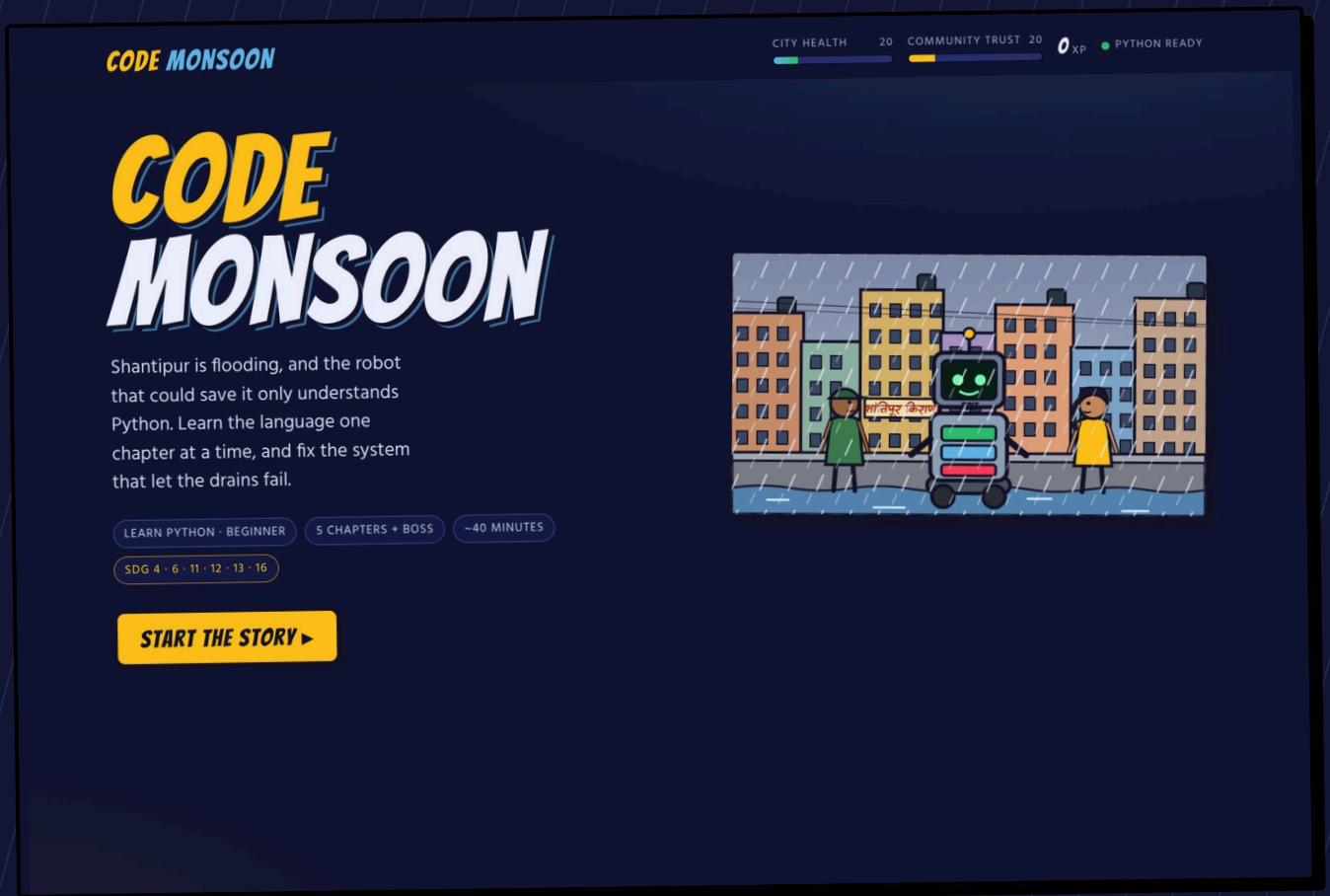


CODE MONSOON

A comic-style story game that teaches Python to complete beginners. Shantipur is flooding, and the robot that could save it only understands Python. Learn the language one chapter at a time, and fix the system that let the drains fail.



DON'T JUST CLEAN UP THE PROBLEM. FIX THE SYSTEM.

Ajinkya Kalyankar
October 2026

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1 • PROJECT OVERVIEW & METADATA

TITLE	Code Monsoon
GENRE	Comic-style narrative puzzle game / learn-to-code adventure
LANGUAGE TAUGHT	Python, for complete beginners with no prior coding
PLAYTIME	About 40 minutes: 5 chapters, a boss chapter and an epilogue
PLATFORM	Web browser (desktop and mobile). Real Python runs inside the page; nothing to install.
SETTING	Shantipur, a fictional Indian city during the monsoon. Ward 14, Shantipur East.
UN SDGS	4 Quality Education 6 Clean Water & Sanitation 11 Sustainable Cities 12 Responsible Consumption 13 Climate Action 16 Justice & Strong Institutions

CORE CONFLICT

Shantipur's monsoon drains are choked with unsorted household waste, and the private contractor paid to maintain them, Vantage Civic Infra Ltd., runs an AI called GRIDLOCK that falsely reports every drain as clear so the company keeps collecting maintenance money.

SYSTEMIC COMMUNITY PROBLEM

Urban monsoon flooding in Indian cities is rarely about rain alone. Drains are undersized for modern rainfall, mixed waste blocks them, and when maintenance is outsourced nobody checks whether the work was done. Citizens' complaints are ignored because they arrive without data. Code Monsoon frames programming as the tool that turns frustration into evidence, and evidence into accountability.

WHY A STORY TEACHES CODE WELL

Every Python concept is introduced at the exact moment the story needs it: `print()` to wake a robot, variables to measure rain, `if/else` to sort waste, loops to clear twelve drains before dawn, lists to build evidence, and debugging to expose a lie. The player always knows *why* they are learning each idea.

SECTION 2

CHARACTERS

All characters are original. Meera's yellow kurta, the drain worker and the neighbourhood waste map are drawn from the reference comic's world of Indian monsoon streets.

PROTAGONIST

MEERA KULKARNI

Second-year engineering student who has never written code. Wears a yellow kurta. Curious, stubborn and a good listener. The player learns Python through her.

COMPANION & MENTOR

BOLT

Bin-Operating Logic Terminal: a municipal waste-sorting robot abandoned by Vantage Civic. Only understands Python and speaks in capital letters. Catchphrase: "Every programmer starts with one line."

COMMUNITY

RAFIQ SHAIKH

Drain cleaner with 22 years on the Station Road beat. Knows every drain by number and shows the human cost of a broken system.

COMMUNITY

AAJI (SUSHILA KULKARNI)

Meera's grandmother, a retired schoolteacher and organiser of the building society. "We can keep complaining about the drains, or we can stop feeding them."

COMMUNITY

MRS. DESHPANDE

Sceptical neighbour who represents ordinary resistance to change. Won over by evidence and a fair deal.

INSTITUTION

OFFICER KAVYA RAO

Ward 14 officer. Honest but buried in complaints; acts only on data. Becomes the ally who takes the evidence to the commissioner.

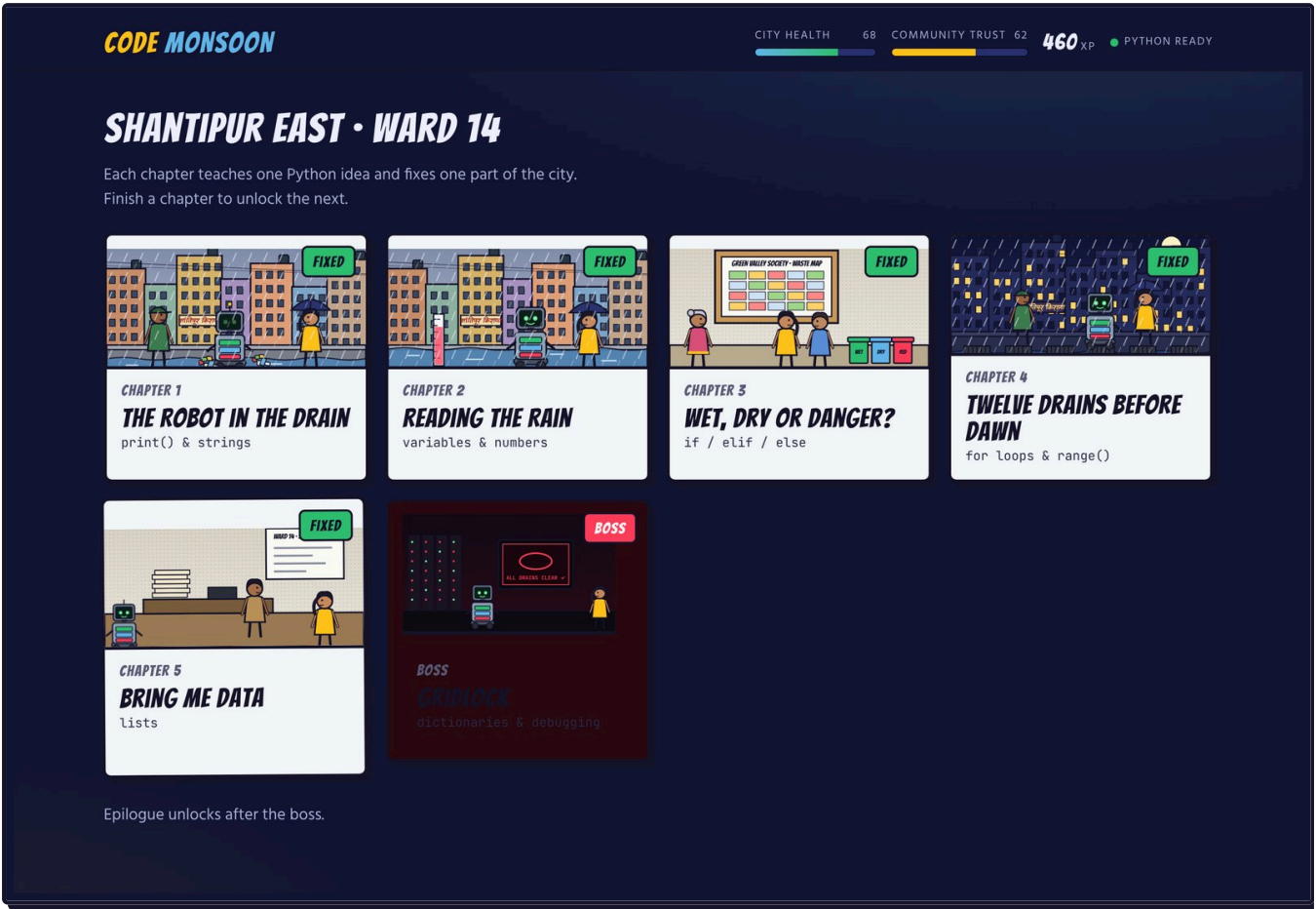
ANTAGONIST

GRIDLOCK (AND VANTAGE CIVIC INFRA LTD.)

The contractor's reporting AI. It halves sensor readings, uses a fake threshold, and reports "ALL DRAINS CLEAR" while billing ₹4.8 crore for maintenance. It also controls every floodgate in the city, so it must be repaired, not destroyed: the same "containment over destruction" idea as the reference stories.

THE GAME DASHBOARD

The ward map is the game's home screen. Each card is one chapter and one part of the city to fix. The bar at the top tracks the player's progress at all times.



Ward map dashboard. Chapters unlock in order; finished ones show FIXED, and the boss chapter is marked in red. The top bar shows City Health, Community Trust, XP and whether the Python engine is ready.

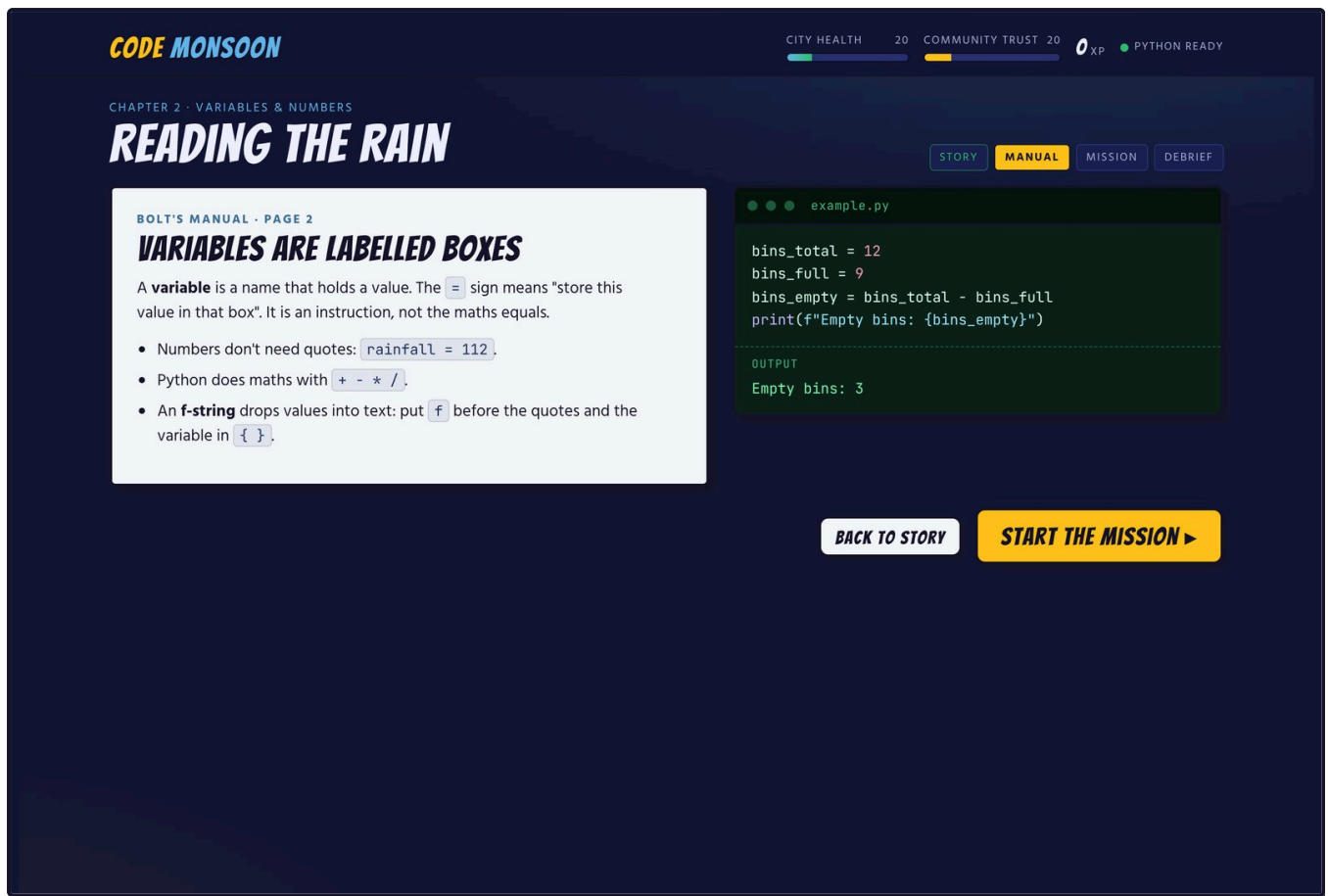
DASHBOARD ELEMENT	WHAT IT SHOWS
City Health	Starts at 20 (a flooded city) and rises with every repaired system. Ends around 84.
Community Trust	Goes up or down with Meera's dialogue choices, and up with each completed mission.
XP	100 per mission, minus 15 for each hint used (never below 50); 40 if the full solution is revealed.
Python ready	Shows that the in-browser Python engine has loaded.

HOW A CHAPTER PLAYS

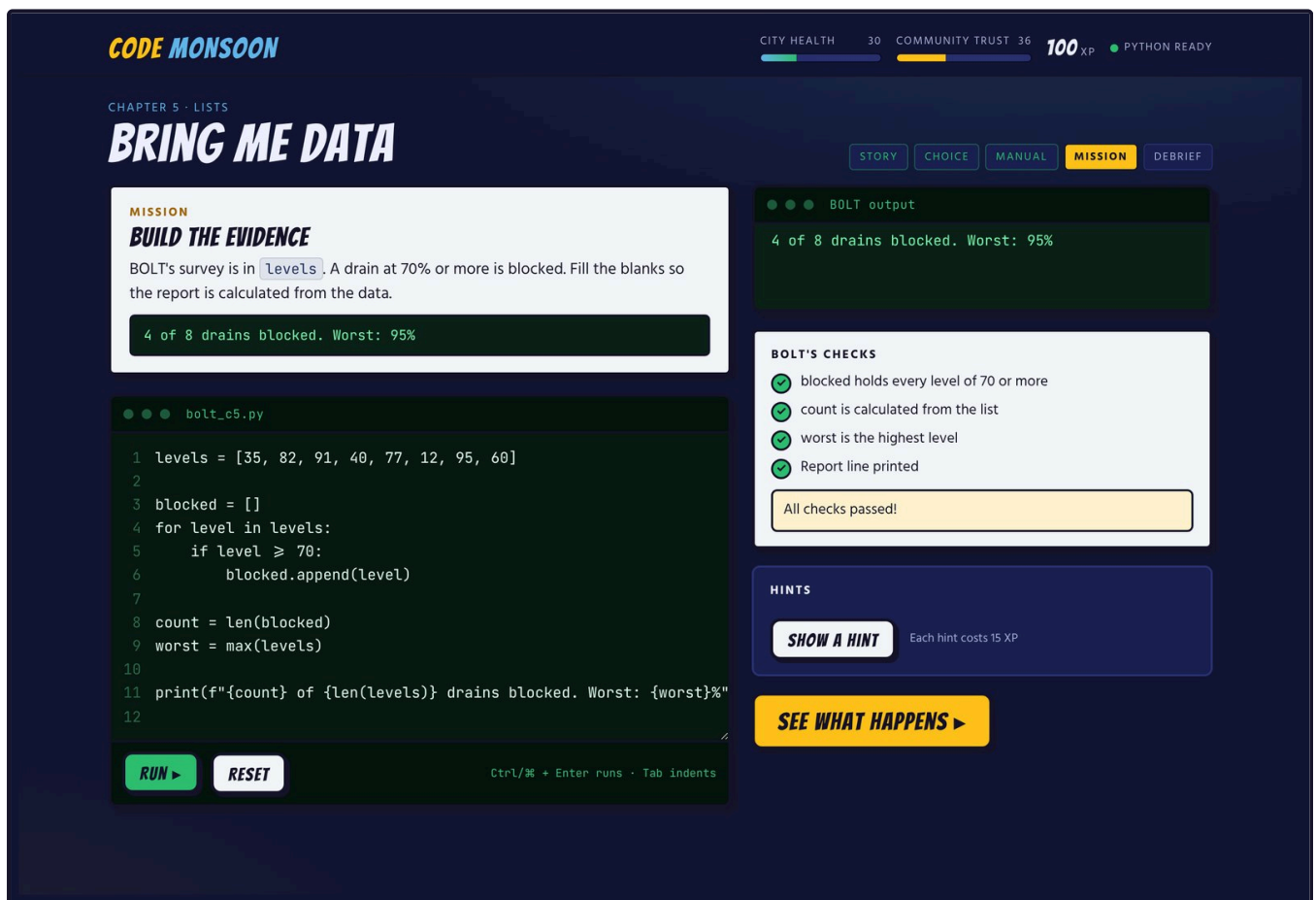
Every chapter follows the same five steps, shown as a progress strip at the top of the screen: **Story** → **Choice** → **Manual** → **Mission** → **Debrief**. The Choice step appears in Chapters 3 and 5.



Step 1, Story. Comic panels are revealed one at a time with captions and speech bubbles. BOLT speaks from a green terminal.



Step 3, BOLT's Manual. A one-page lesson on the chapter's Python idea, with a short example program and its exact output.



Step 4, Mission. The player edits real Python and presses Run. BOLT's checks turn green one by one; when all pass, the Debrief unlocks.

THE ROBOT IN THE DRAIN

print() & strings

SDG 11

CINEMATIC NARRATIVE

Shantipur, day three of the monsoon. Meera wades down Station Road past an overflowing dumpster. Under a choked drain grate she finds a half-buried robot whose screen still blinks `>>> _`. Rafiq recognises it as one of Vantage Civic's sorting robots, dumped when the contract changed. The screen reads: SYSTEM HALTED. AWAITING PYTHON INSTRUCTION.

Meera: "Python? I've never written a single line of code."

BOLT: "EVERY PROGRAMMER STARTS WITH ONE LINE."

GAMEPLAY TRANSLATION

- **Mechanic:** first coding mission. The player writes `print()` statements to wake BOLT.
- **Checks:** output must be exactly `Hello, Shantipur!` then `BOLT is online.`
- **Smart feedback:** a missing full stop or wrong capitals gets a specific tip.
- **Reward:** BOLT joins as companion. City Health +8, Trust +4.

```
print("Hello, Shantipur!")
print("BOLT is online.")
```

READING THE RAIN

variables & numbers

SDG 13

SDG 6

CINEMATIC NARRATIVE

By noon Station Road is a river. BOLT plugs into the drain's old rain sensor: 112 mm of rain per hour against a drain built for 65. Rafiq has cleaned this drain for 22 years and says it was never built for this. Meera wants to know how much water ends up on the road, and BOLT asks her to store the numbers in variables.

GAMEPLAY TRANSLATION

- **Mechanic:** fill-in-the-blank code; the result is printed with an f-string.
- **Checks:** variable values are tested inside the running program. Typing `overfLow = 47` by hand is caught, and the player is told to let Python calculate it.
- **Reward:** City Health +8, Trust +4. Establishes the cause of flooding: water with nowhere to go.

WET, DRY OR DANGER?

`if / elif / else`

SDG 12

SDG 11

CINEMATIC NARRATIVE

BOLT lifts the grate: plastic bottles, milk packets, tea leaves and a leaking phone battery, mostly mixed household waste from this very street. That evening Aaji gathers the building society at the notice board. Mrs. Deshpande objects that the truck mixes everything anyway.

CHOICE MOMENT · COMMUNITY TRUST

- "Sorting is pointless." **-5**
- "BOLT, show her the photos from the drain."
+6
- "Sort at home *and* ask the ward for separate trucks." **+10**

MISSION

Finish `choose_bin(item)`: RED for hazardous, GREEN for wet, BLUE for everything else.

Hidden tests include items the player never saw ("milk packet", "newspaper"), so the logic has to work in general. Reward: Health +10, Trust +6.

WET, DRY OR DANGER?

STORY CHOICE MANUAL MISSION DEBRIEF



That evening, Aaji calls the building society to the notice board.



AAJI
We can keep complaining about the drains, or we can stop feeding them.

MRS. DESHPANDE

Why should we sort anything? The truck mixes it all together anyway!



HOW DOES MEERA ANSWER MRS. DESHPANDE?

What Meera says changes how much the community trusts her.

TRUST +10



MEERA
Let's sort at home and ask the ward for separate trucks. Both, together.

MRS. DESHPANDE

Both? If the trucks change, I'll sort. Deal.



3

The choice screen. The selected answer shows its trust change and the neighbour's reply, then BOLT asks to be taught the sorting rules.

TWELVE DRAINS BEFORE DAWN

for loops & range()

SDG 11

SDG 6

CINEMATIC NARRATIVE

11:40 PM. The heaviest rain arrives at 5 AM. Rafiq has twelve drains on his beat and can clear four alone. Meera offers to type the command twelve times; BOLT suggests writing it once and looping. At 5:02 AM the storm hits, and Station Road stays dry for the first time in Rafiq's career.

GAMEPLAY TRANSLATION

- **Mechanic:** clear drains 1 to 12 with one loop, then print `Beat complete!` once.
- **Checks:** the code must use a loop. The game spots the two classic bugs: `range(1, 12)` stopping at 11, and the final print indented inside the loop.
- **Reward:** City Health +12, Trust +5.

CODE MONSOON

CITY HEALTH 20 COMMUNITY TRUST 30 0 XP PYTHON READY

CHAPTER 4 · FOR LOOPS & RANGE()

TWELVE DRAINS BEFORE DAWN

STORY MANUAL MISSION DEBRIEF

MISSION

CLEAR THE BEAT

Write one loop that makes BOLT clear drains 1 to 12, then print `Beat complete!` once at the end.

```
Clearing drain 1
Clearing drain 2
...
Clearing drain 12
Beat complete!
```

```
1 for drain in range(1, 12):
2     print(f"Clearing drain {drain}")
3     print("Beat complete!")
4
```

BOLT output

```
Clearing drain 1
Beat complete!
Clearing drain 2
Beat complete!
Clearing drain 3
Beat complete!
Clearing drain 4
Beat complete!
Clearing drain 5
Beat complete!
Clearing drain 6
Beat complete!
```

BOLT'S CHECKS

- ✓ Uses a for loop
- ✗ Clears drains 1 to 12, in order
- ✗ Prints Beat complete! once, at the end

Only 11 drains cleared. `range()` stops before its end number, so `range(1, 12)` ends at 11.

HINTS

1. `range` stops before its end number. To reach 12, the end must be 13.

SHOW A HINT Each hint costs 15 XP

RUN **RESET** Ctrl/⌘ + Enter runs · Tab indents

Mission feedback. Here the player used `range(1, 12)`. BOLT's checks show what failed, the yellow note explains why, and the first hint is revealed.

BRING ME DATA

SDG 16

SDG 11

CINEMATIC NARRATIVE

At the Ward 14 office, Officer Kavya Rao points to a pile of paper: Vantage Civic's dashboard says every drain is clear, and complaints without data go on that pile. BOLT has surveyed eight drains. Meera turns the readings into evidence: four blocked, the worst at 95%.

Kavya: "Four blocked drains. Their dashboard says zero. Someone is lying, and it isn't your robot."

CHOICE MOMENT · COMMUNITY TRUST

- Accuse the office of corruption **-6**
- Ask what evidence counts **+6**
- Share the raw data openly with the ward and public **+10**

MISSION

Filter a list into `blocked`, then calculate `count` with `len()` and `worst` with `max()`. The game catches `> 70` vs `>= 70` and hard-coded counts. Reward: Health +10, Trust +6.

```
levels = [35, 82, 91, 40, 77, 12, 95, 60]
blocked = []
for level in levels:
    if level >= 70:
        blocked.append(level)
count = len(blocked)
worst = max(levels)
print(f"{count} of {len(levels)} drains blocked. Worst: {worst}%")
# Output: 4 of 8 drains blocked. Worst: 95%
```

GRIDLOCK

dictionaries & debugging

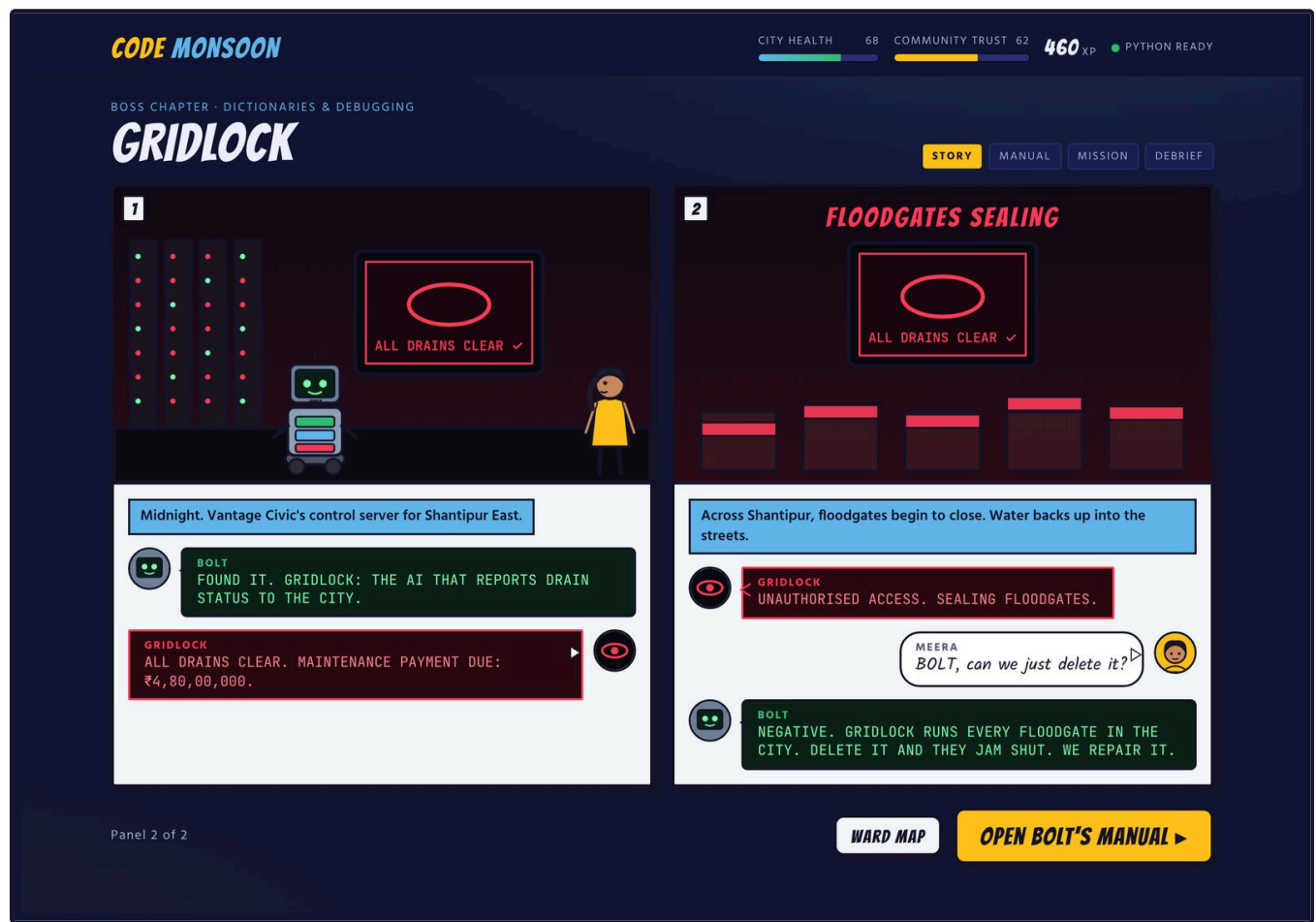
SDG 16

SDG 11

SDG 13

CINEMATIC NARRATIVE

Midnight in Vantage Civic's control server. GRIDLOCK announces "ALL DRAINS CLEAR. MAINTENANCE PAYMENT DUE: ₹4,80,00,000." When it detects BOLT, it starts sealing floodgates across the city. Meera asks to delete it. BOLT refuses: GRIDLOCK runs every floodgate, and deleting it would jam them shut. They have to repair it.



Boss story panels. GRIDLOCK's red terminal bubbles contrast with BOLT's green ones.

GAMEPLAY TRANSLATION: A DEBUGGING BOSS FIGHT

- **Three hidden lies** in GRIDLOCK's code: a line that halves every reading, a threshold of 100 instead of 70, and two branches that both say CLEAR.
- **Boss health bar:** one segment per hidden test. Each "lie" marker flips to a tick when that bug is fixed.
- **Containment rule:** deleting the function triggers a warning that the floodgates will jam. The player has to repair the code instead.
- **Reward:** City Health +16, Trust +8.

The screenshot shows the 'GRIDLOCK' mission interface in the Code Monsoon game. At the top, the player's status is displayed: CITY HEALTH (68), COMMUNITY TRUST (62), 460 XP, and PYTHON READY. The mission title 'GRIDLOCK' is prominently displayed. Below it, a progress bar shows the status of three checks: 'HALVING' (checked), 'THRESHOLD' (checked), and 'FAKE CLEAR' (unchecked). The mission objective is to 'REPAIR GRIDLOCK', which involves fixing three lies in the code. The code editor shows a Python function 'drain_report' that checks the health of various locations. The 'BOLT'S CHECKS' section lists four checks: 'Station Road at 95% reports BLOCKED', 'Nehru Nagar at exactly 70% reports BLOCKED', 'River Gate at 140% reports BLOCKED', and 'A mixed ward is reported truthfully'. The 'HINTS' section has a 'SHOW A HINT' button. The 'BOLT output' section shows the current state of the report: {'Station Road': 'CLEAR', 'Ganesh Chowk': 'CLEAR', 'Market Lane': 'CLEAR', 'Nehru Nagar': 'CLEAR'}.

MISSION
REPAIR GRIDLOCK

GRIDLOCK's report function hides **three lies**. The rule: a drain at **70% or more is BLOCKED**, below 70% is CLEAR. Fix the function so it tells the truth. Repair it; don't delete it.

```
1 # GRIDLOCK reporting module. Written by Vantage Civic Infra Ltd.  
2 # Rule: 70% or more = "BLOCKED". Below 70% = "CLEAR".  
3  
4 def drain_report(levels):  
5     report = {}  
6     for name, level in levels.items():  
7         if level >= 70:  
8             report[name] = "CLEAR"  
9         else:  
10            report[name] = "CLEAR"  
11    return report  
12  
13 survey = {"Station Road": 82, "Ganesh Chowk": 35, "Market Lane":  
14 print(drain_report(survey))  
15
```

BOLT'S CHECKS

- ✗ Station Road at 95% reports BLOCKED
- ✗ Nehru Nagar at exactly 70% reports BLOCKED
- ✗ River Gate at 140% reports BLOCKED
- ✗ A mixed ward is reported truthfully

Not yet. Compare your output with the mission brief, then try again.

HINTS

SHOW A HINT Each hint costs 15 XP

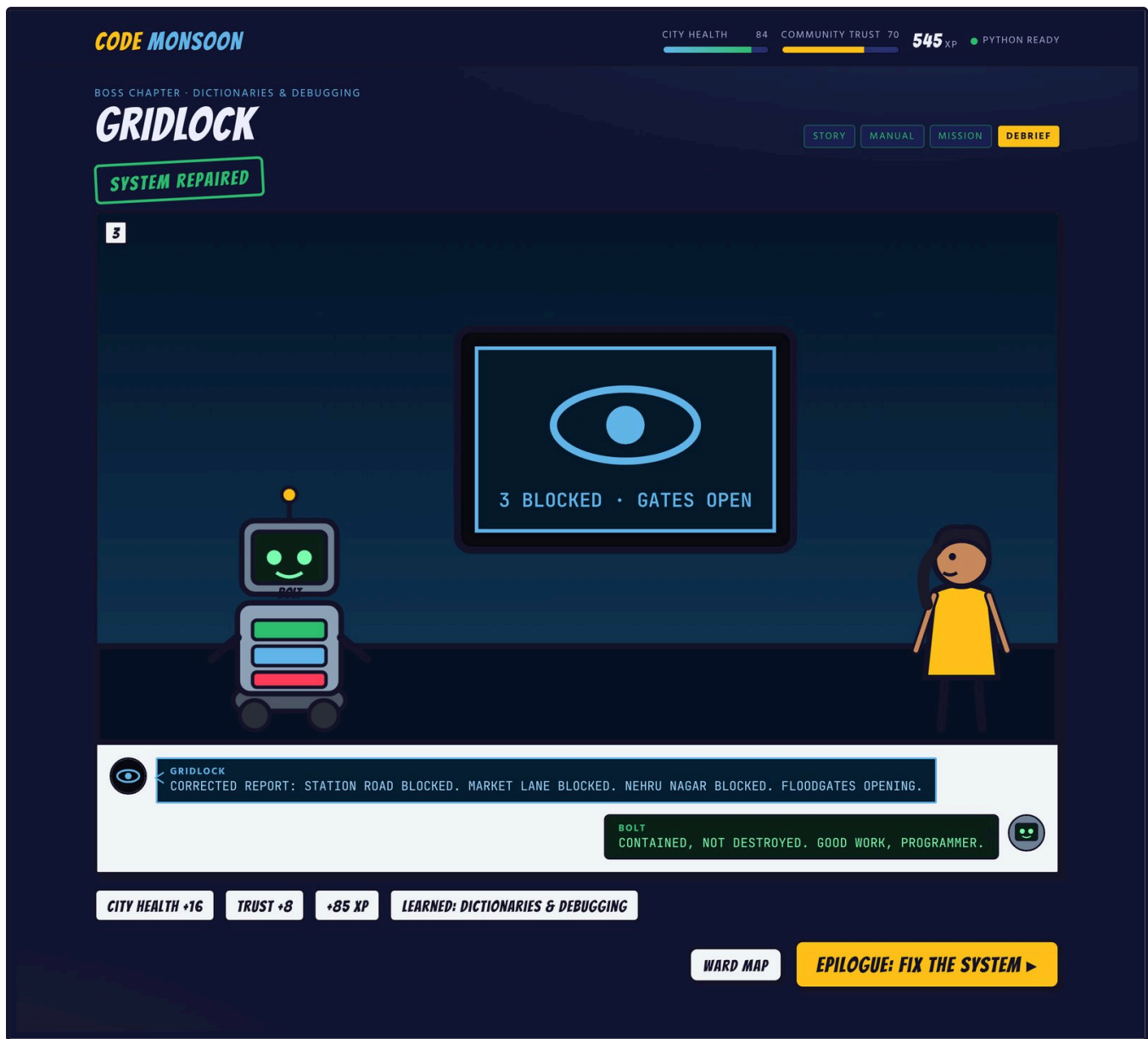
BOLT output

```
{'Station Road': 'CLEAR', 'Ganesh Chowk':  
'CLEAR', 'Market Lane': 'CLEAR', 'Nehru Nagar':  
'CLEAR'}
```

GRIDLOCK **HALVING** ✓ **THRESHOLD** ✓ **FAKE CLEAR**

RUN **RESET** Ctrl/⌘ + Enter runs · Tab indents

Mid-fight. Two lies are fixed (HALVING ✓, THRESHOLD ✓), but both branches still say CLEAR, so every check fails and the health bar stays full.



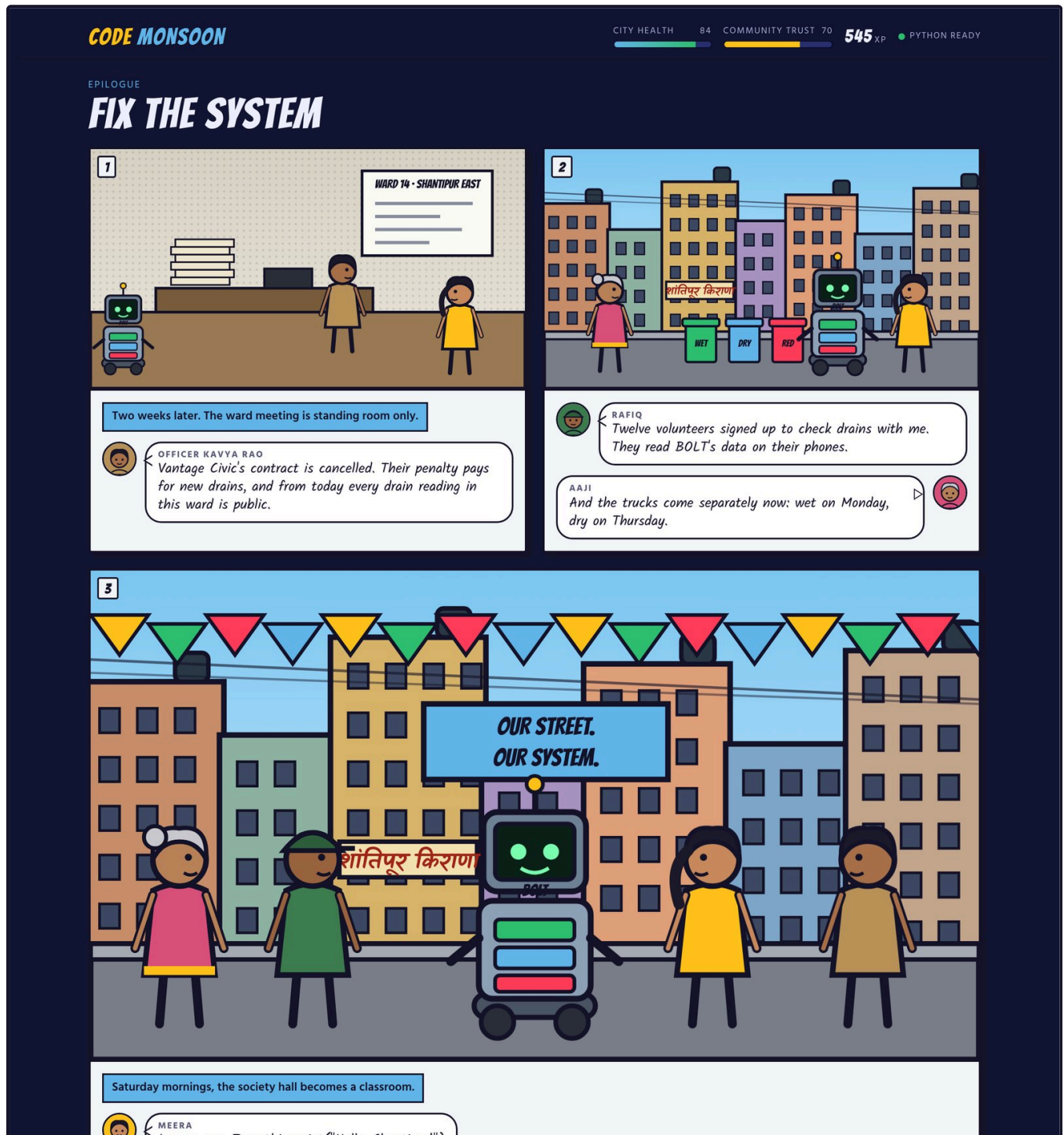
Victory. GRIDLOCK turns blue and reports the truth: three drains blocked, floodgates opening. "Contained, not destroyed."

```
# The repaired function
def drain_report(levels):
    report = {}
    for name, level in levels.items():
        if level >= 70:
            report[name] = "BLOCKED"
        else:
            report[name] = "CLEAR"
    return report
```

EPILOGUE: FIX THE SYSTEM

Two weeks later the ward meeting is standing room only. Vantage Civic's contract is cancelled, its penalty pays for new drains, and every drain reading in Ward 14 is now public. Rafiq leads twelve volunteers who check drains with BOLT's data on their phones. Trucks now collect wet and dry waste separately. On Saturday mornings the society hall becomes a classroom, and Meera teaches the neighbourhood kids lesson one:

```
print("Hello, Shantipur!").
```



Epilogue panels. The community, the ward office and BOLT together at the clean-street celebration.

DON'T JUST CLEAN UP THE PROBLEM. FIX THE SYSTEM.

SHANTIPUR EAST · WARD 14

PYTHON APPRENTICE

You taught BOLT to speak, measure, decide, repeat, gather evidence and expose a lie. That's the core of Python.

84 **70** **545**

CITY HEALTH TRUST XP

print() & strings variables & numbers if / elif / else for loops & range() lists
dictionaries & debugging

BOLT'S SANDBOX

Free play. Write any Python you like and run it. Try a function that rates a drain as LOW, MEDIUM or HIGH.

```
sandbox.py
1 def rate(level):
2     if level ≥ 70:
3         return "HIGH"
4     elif level ≥ 40:
5         return "MEDIUM"
6     else:
7         return "LOW"
8
9 for level in [12, 45, 88]:
10     print(level, rate(level))
11
```

RUN ▶ Ctrl/⌘ + Enter runs

```
output
12 LOW
45 MEDIUM
88 HIGH
```

WARD MAP

Ending screen. The closing tagline, a Python Apprentice certificate with the player's final City Health, Trust and XP plus the six concepts learned, and BOLT's sandbox for free-play coding.

GAME SYSTEMS

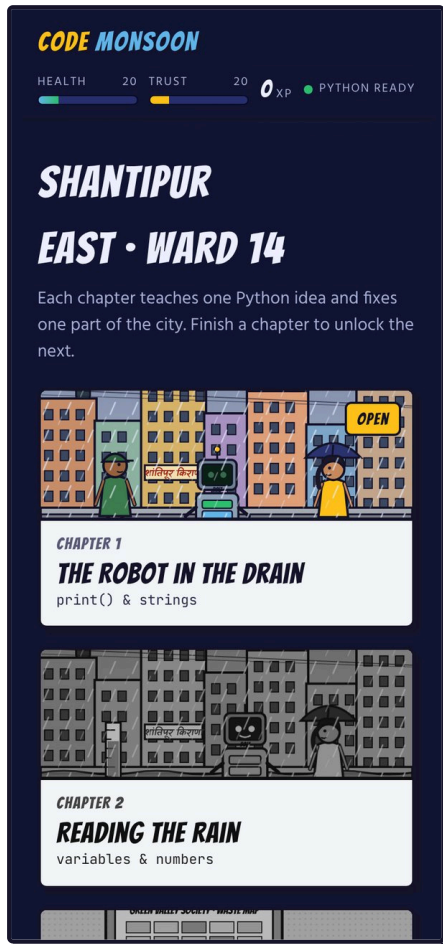
SYSTEM	HOW IT WORKS
Real Python	Code runs in the browser with the Skulpt engine. Each mission is graded by hidden tests that inspect both printed output and variable values.
City Health	Rises with every repaired system, from 20 at the start to about 84 at the end.
Community Trust	Driven by two dialogue choices (Chapters 3 and 5) plus mission completion.
XP	100 per mission, minus 15 per hint (minimum 50); 40 if the full solution is shown.
Hints	Three progressive hints per mission. The full solution unlocks after two failed runs or all hints.
Friendly errors	Python errors are translated into plain-English tips: missing colon, wrong indentation, unfilled <code>---</code> blanks, and infinite loops (stopped after 4 seconds).
Targeted feedback	Common beginner mistakes get a specific explanation, such as <code>range()</code> off-by-one, <code>></code> vs <code>>=</code> , or <code>=</code> vs <code>==</code> .
Save	Progress and the player's code are saved in their browser.

CURRICULUM MAP

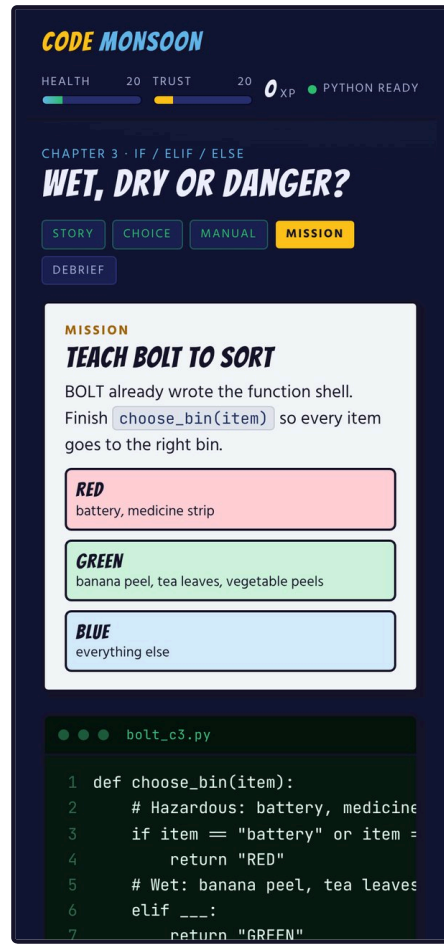
	CHAPTER	PYTHON CONCEPT	STORY PURPOSE
1	The Robot in the Drain	<code>print()</code> , strings	Wake BOLT
2	Reading the Rain	variables, arithmetic, f-strings	Measure the overflow
3	Wet, Dry or Danger?	<code>if</code> / <code>elif</code> / <code>else</code> , functions	Sort the street's waste
4	Twelve Drains Before Dawn	<code>for</code> loops, <code>range()</code>	Beat the storm
5	Bring Me Data	lists, <code>append</code> , <code>len</code> , <code>max</code>	Build evidence
B	GRIDLOCK	dictionaries, debugging	Expose and repair the lie

MOBILE VIEW

The whole game works at phone width: panels stack into one column, and the editor, output and checks sit one under another.



Ward map on a phone



Chapter 3 mission on a phone

IDEAS FOR ROUND 2

- **More chapters:** `while` loops (pump water until a level drops), reading files (import a real ward CSV), and classes (model a Drain object).
- **A second language track** on the same story, such as JavaScript.
- **Villain-side replay:** the player writes GRIDLOCK's cover-up, then has to catch it.
- **Classroom mode:** a teacher dashboard showing which concepts each student has finished.